

# Huynh Trinh Ngoc

Medical Image Analysis, Computer Vision, Generative AI

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## EDUCATION

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**VNU University of Engineering and Technology**  
*M.Sc. Candidate (Expected 06/2026) in Computer Science*  
CGPA: 3.78/4.00

**Hanoi, Vietnam**  
09/2024 – Present

**VNU University of Engineering and Technology**  
*B.Sc. in Information Technology*  
CGPA: 3.86/4.00

**Hanoi, Vietnam**  
09/2020 – 06/2024

**High School for Gifted Students, VNU University of Science**  
*High School Diploma in Mathematics*  
CGPA: 9.4/10.0

**Hanoi, Vietnam**  
09/2017 – 06/2020

## WORK EXPERIENCE

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**Institute for Artificial Intelligence, VNU-UET**  
*Teaching Assistant & M.Sc. Student*

**Hanoi, Vietnam**  
09/2024 – Present

**Courses assisted:** Introduction to Programming; Relational Databases and SQL; Computational Thinking; Programming Methodology; Practical Artificial Intelligence Systems Development.

**Research interests:** Research on medical image analysis leveraging generative methods (e.g., VAE, diffusion, and flow matching), focusing on uncertainty-aware segmentation, weakly supervised anomaly detection, and interpretable prediction of cognitive disorders.

## ONGOING RESEARCH

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**Latent Representation Learning from 3D Brain MRI for Interpretable Prediction in MS**

*Supervisors: Tran Quoc Long, Nguyen Linh Trung, Guy Nagels*

06/2025 – Present

- Develop InfoVAE-Med3D for interpretable latent representation learning from 3D brain MRI.
- Learn structured latent spaces for brain age and cognitive impairment prediction in multiple sclerosis.
- Improve latent representation quality through mutual information maximization for downstream analysis.
- Manuscript submitted to Neuroinformatics in 11/2025 and currently under review.

**An Efficient Diffusion-Based Weakly Supervised Anomaly Detection in Medical Images**

*Supervisor: Tran Quoc Long*

10/2025 – Present

- Develop a diffusion-based framework for weakly supervised medical anomaly detection.
- Reconstruct healthy counterparts from abnormal medical images using guided diffusion sampling.
- Design a latent-space diffusion framework with VQ-VAE for faster and more efficient inference..
- Complete the manuscript and prepare for journal submission.

## PUBLICATIONS

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- Aleatoric uncertainty estimation in medical image segmentation via flow matching. Van Nguyen Phi, **Huynh Trinh Ngoc**, Duy Minh Lam Nguyen, Phu Loc Nguyen, Quoc Long Tran. 2025. In: *International Workshop on Uncertainty for Safe Utilization of Machine Learning in Medical Imaging (UNSURE Workshop at MICCAI)*, pp. 134–144. Springer Nature Switzerland. DOI: [10.1007/978-3-032-06593-3\\_13](https://doi.org/10.1007/978-3-032-06593-3_13).

- LatentFM: A latent flow matching approach for generative medical image segmentation. **Huynh Trinh Ngoc**, Hoang Anh Nguyen Kim, Toan Nguyen Hai, Long Tran Quoc. 2026. In: *FISU Joint Conference on Artificial Intelligence (FJCAI 2026)*, Can Tho, Vietnam. Preprint: [arXiv:2512.04821](https://arxiv.org/abs/2512.04821).
- Diffusion model in latent space for medical image segmentation. **Huynh Trinh Ngoc**, Toan Nguyen Hai, Ba Luong Son, Long Tran Quoc. 2026. In: *FISU Joint Conference on Artificial Intelligence (FJCAI 2026)*, Can Tho, Vietnam. Preprint: [arXiv:2512.01292](https://arxiv.org/abs/2512.01292).
- The Privacy–Utility Trade-off in Brain MRI Synthesis: A Comparative Framework for Generative Models. Dam Thai Ninh, Nguyen Duc Kien, **Trinh Ngoc Huynh**, Le Quoc Anh, Dinh Tran Hiep, Bui Duc Manh, Nguyen Hai Anh, Miguel D’Haeseleer, Jeroen Van Schependom, Stijn Denissen, Tran Quoc Long, Nguyen Linh Trung, and Guy Nagels. 2025. In: *14th International Symposium on Information and Communication Technology (SOICT)*, Nha Trang, Vietnam, December 2025.

## PROJECTS

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**Ministry of Science and Technology**

*Research Project Member*

**Hanoi, Vietnam**

*08/2022 – 11/2025*

**Title:** Research on the Application of Artificial Intelligence in Lung Cancer Diagnosis through Analysis of Chest CT, Bronchoscopy, and Histopathological Images

**Role:** Member

**Status:** Completed

## AWARDS AND HONORS

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- Viettel Research Excellence Scholarship, 2025
- Full scholarship in 6/8 semesters, 2020-2024.
- 3rd prize in the *Scientific Research Conference for Undergraduates at FIT-UET* with project Diffusion model in latent space for medical image segmentation, 2024.
- Consolation prize in the *Scientific Research Conference for Undergraduates at FIT-UET* with project Developing a GPU-Optimized Real-time AI Surveillance System for Camera, 2024.
- 3rd prize in the *Scientific Research Conference for Undergraduates at FIT-UET* with project Multi-tracking multi-camera project, 2023.
- EVN Scholarship for VNU Students, Academic Year 2022-2023
- Top 5 in the *Air Quality Forecasting Challenge (AI4VN)*, 2022

## LANGUAGES

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- **Vietnamese:** Native
- **English**

## REFERENCES

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- **Dr. Long Tran Quoc:** Head of the Institute of Artificial Intelligence, University of Engineering and Technology, Vietnam National University  
Contact: [tqlong@vnu.edu.vn](mailto:tqlong@vnu.edu.vn)
- **Prof. Dr. Nguyen Linh Trung:** Signal and Systems Laboratory, Faculty of Electronics and Telecommunications, University of Engineering and Technology, Vietnam National University  
Contact: [lingtrung@vnu.edu.vn](mailto:lingtrung@vnu.edu.vn)
- **Prof. Dr. Ir. Guy Nagels:** Chair of the AIMS Research Group, Vrije Universiteit Brussel  
Contact: [guy.nagels@vub.be](mailto:guy.nagels@vub.be)